BUDDY: Bimanual Underwater Demonstration-Driven Policy

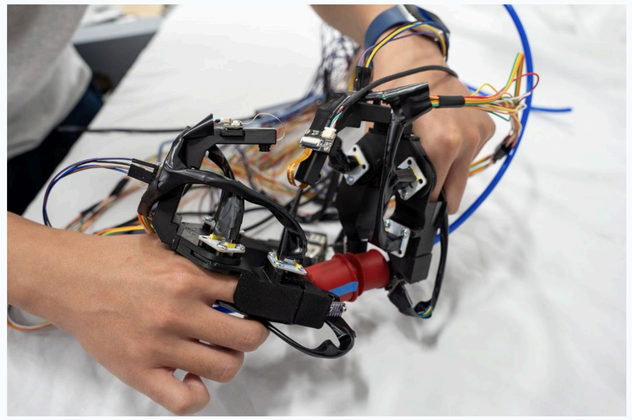
Soyoun Choi[1], Vivek Patel[1], Lukas Klostermair[1], Juhyun Jung[1], Hannah Watkins[1], Oussama Khatib[1], Mark Cutkosky[1]

[1]Mechanical Engineering Department, Stanford University

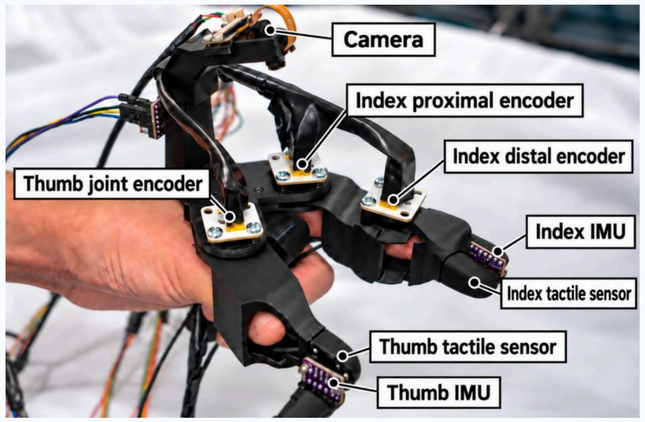
Introduction

Underwater manipulation changes contact, perception, and motion.
We collect synchronized bimanual demonstrations to identify the minimum hand dexterity and sensing needed for reliable manipulation.

Demonstration, sensing and robot hardware



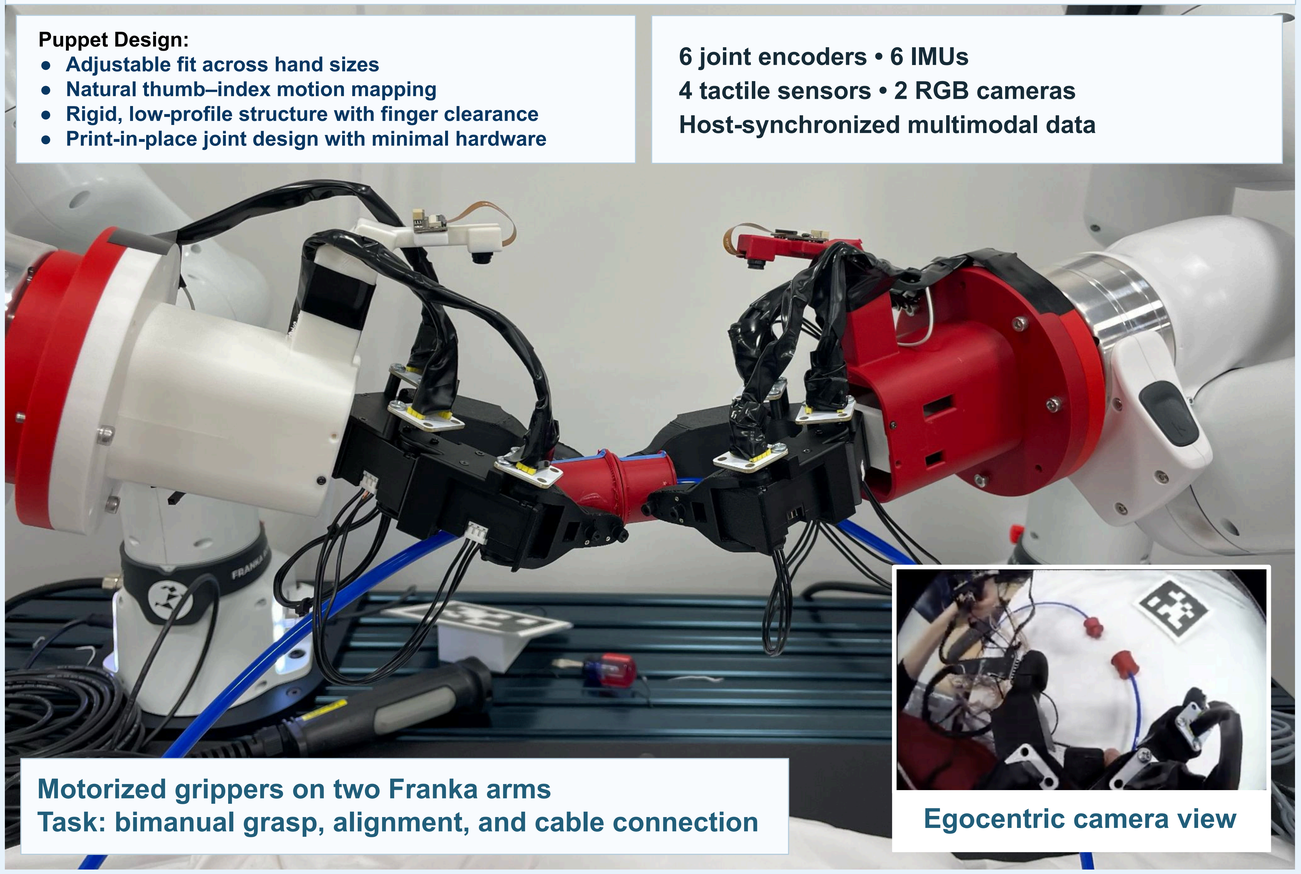
Human demonstrations



Instrumented demonstration gripper

Puppet Design:
- Adjustable fit across hand sizes
- Natural thumb–index motion mapping
- Rigid, low-profile structure with finger clearance
- Print-in-place joint design with minimal hardware

6 joint encoders • 6 IMUs
4 tactile sensors • 2 RGB cameras
Host-synchronized multimodal data



Motorized grippers on two Franka arms
Task: bimanual grasp, alignment, and cable connection

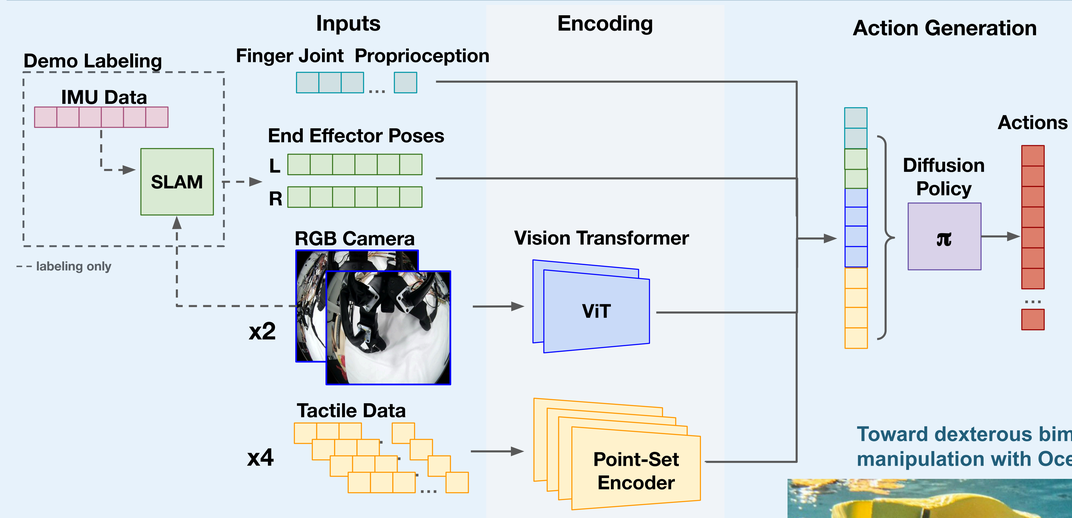
Egocentric camera view

Manipulation in air vs. underwater

FACTOR	IN AIR	UNDERWATER
DYNAMICS	Fluid loads often small	Drag, buoyancy and added mass affect motion Currents can disturb the object
VISION	No water-induced optical effects	Absorption and scattering reduce visibility Refraction changes geometry through camera ports
CONTACT	Dry friction depends on the interface	Water changes friction and separation forces Effects depend on materials, load and motion
HARDWARE	No seawater or depth-related pressure	Depth pressure, corrosion and sealing constrain design Protect motors, sensors and transmissions
SENSING	Standard sensor packaging and calibration	Force/tactile and acoustic sensing can complement vision Sensors need suitable packaging and calibration

Goal: achieve useful underwater dexterity with minimal hand complexity and the smallest sensing set that preserves task performance.

Learning across embodiments



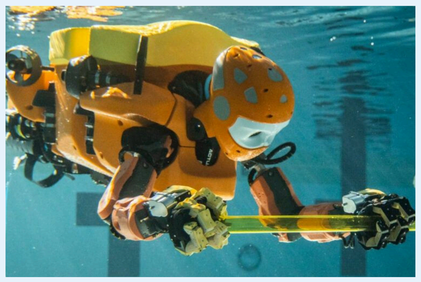
Input ablation: RGB → RGB + proprioception → RGB + proprioception + tactile

Goal: bridge the human → robot → underwater embodiment gap

Next steps

Model water-aware dynamics in simulation
Ablate policy inputs to identify necessary sensing
Evaluate in-air → underwater policy transfer
Co-design the Ocean5K hand and pressure-compensated glove

Toward dexterous bimanual manipulation with Ocean5K



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The Stanford Robotics Center

References: [1] Sivcev et al., Ocean Engineering, 2018; [2] Khatib et al., IEEE RAM, 2016; [3] Stuart et al., IJRR, 2017; [4] Todorov et al., IROS, 2012; [5] Li et al., arXiv, 2026; [6] Kampmann & Kirchner, RAS, 2015.